NEEVETHA N

Software Systems Student | AI, Machine Learning & Intelligent Systems

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PROFESSIONAL SUMMARY

Software Systems student (CGPA 8.89/10) specializing in Machine Learning, Multimodal Deep Learning, and Intelligent Systems. Experienced in designing and evaluating end-to-end ML pipelines across computer vision, behavioral analytics, and NLP - spanning frequency-domain deepfake detection, civic-intelligence policy verification, and RAG architectures - supported by national hackathon recognition and research internships.

EDUCATION

Coimbatore Institute of Technology (CIT) - M.Sc. Software Systems (5-Year Integrated) 2024 - 2029 | Coimbatore, India
Academic Standing: CGPA 8.89 / 10.0 | Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Computer Networks, Operating Systems, Web Technologies

TECHNICAL SKILLS

Languages: Python, C++, C, SQL, Java

Machine Learning: Scikit-learn, XGBoost, Random Forest, Feature Engineering, Association Rule Mining (Apriori, FP-Growth), SHAP

Deep Learning & NLP: PyTorch, Hugging Face (LoRA, Transformers), IndicBERT, Sentence-BERT, EfficientNet, BiLSTM, Wav2Vec2, FAISS, RAG, NER

Data Analysis & Viz: Pandas, NumPy, Matplotlib, Seaborn, Power BI, Excel

Databases & Storage: PostgreSQL, Supabase, MySQL, SQLite, Firebase

MLOps & Backend Tools: FastAPI, Docker, MLflow, Git, GitHub, Streamlit, Linux

FEATURED MACHINE LEARNING & AI PROJECTS

AEGIS - *Multi-Modal Deepfake Detection System* [github.com/niviii-08/aegis]

Tech Stack: Python, PyTorch, FastAPI, OpenCV, Pandas, NumPy, EfficientNet-B4, BiLSTM, Wav2Vec2, FFT Fusion, MLflow

- **Multimodal Fusion Pipeline:** Engineered an end-to-end deepfake detection system integrating EfficientNet-B4 (spatial visual artifacts), BiLSTM (temporal frame incoherence), and Wav2Vec2 (synthetic voice forensics).
- **Frequency-Domain Analysis:** Integrated 2D FFT spectral feature extraction and cross-modal attention fusion to expose subtle blending boundaries overlooked by standard RGB networks.
- **Reproducible MLOps & Explainability:** Tracked training runs and model calibration via MLflow; integrated Grad-CAM saliency heatmaps to provide visual explainability for high-stakes verification.

CivicLens TN - *AI-Powered Public Accountability & Policy Intelligence Platform*

Tech Stack: Python, FastAPI, IndicBERT, Sentence-BERT, XGBoost, Isolation Forest, K-Means, DBSCAN, Scikit-learn, Streamlit

- **Semantic Claim Verification:** Built an evidence-based policy verification module leveraging fine-tuned IndicBERT and Sentence-BERT to cross-reference political promises against legislative records.
- **Anomaly Detection & Clustering:** Applied Isolation Forest and XGBoost to identify statistical spending anomalies in public budget allocations; grouped related state welfare schemes using K-Means and DBSCAN.
- **Media Framing Analysis:** Extracted cross-outlet news bias and narrative framing using TF-IDF and transformer embeddings across regional Tamil news transcripts.

ContextIQ - *Personal Behaviour Intelligence & Cognitive Friction Reduction System*

Tech Stack: Python, FastAPI, PostgreSQL, Next.js 14, XGBoost, Random Forest, SHAP, Association Rule Mining (FP-Growth)

- **Behavioral Association Mining:** Applied Apriori and FP-Growth algorithms to discover recurring app-switching patterns and contextual triggers leading to cognitive interruption.
- **Predictive Cost Modeling:** Trained XGBoost and Random Forest regression models to predict task-resumption recovery latency with leakage-safe walk-forward validation.
- **Explainability & Service Deployment:** Interpreted feature contributions using SHAP summary values; served inferences via modular FastAPI endpoints connected to a Next.js 14 dashboard.

LifeLink - *Smart Blood Donation & Intelligent Matching Platform* [github.com/niviii-08/Lifelink]

Tech Stack: Python, FastAPI, PyTorch, DistilBERT, Hugging Face (LoRA, RAG), FAISS Vector Search, Leaflet, Firebase

- **Medical NLP & Urgency Triage:** Fine-tuned DistilBERT with LoRA for Named Entity Recognition (NER) to automatically extract blood groups, unit quantities, and hospital locations from unstructured text.

- **FAISS-Backed Semantic Retrieval:** Implemented a Retrieval-Augmented Generation (RAG) pipeline with FAISS vector search to match emergency donor requests against inventory in real time.
- **Demand Forecasting:** Trained predictive regression models to forecast regional seasonal blood shortages, optimizing donor outreach schedules.

PROFESSIONAL EXPERIENCE

Funobotz Paper Robotics Pvt. Ltd. - Online Research Intern (via Praya Labs) May 2026 - Jun 2026 | Remote

Credential: Certificate No. INPLNE026 | Focus: Conversational AI & Educational Chatbot Systems

- **Conversational Logic Architecture:** Contributed to the core conversational dialogue trees and intent routing logic of the Funobotz AI Chatbot System across the development lifecycle.
- **Validation & Quality Assurance:** Collaborated with engineering leads on automated API testing, dialogue edge-case verification, and system validation to ensure high intent classification reliability.

HONORS & HACKATHON ACHIEVEMENTS

- **RAC-A-THON 2025:** Awarded Best Theme & Special Mention in a 24-hour national-level hackathon organized by Rotaract Club.
- **Codeathon 2.0 (Thiran 2026):** Selected for an Internship Opportunity through competitive project evaluation at the national-level 24-hour hackathon.
- **Tech Sprint 2025:** Recognized with Special Mention for innovative architectural solution design.
- **Technical Competitions:** Active participant and finalist at Melinia 2026 and Intellina 2026 technical symposiums.

CERTIFICATIONS & LEADERSHIP

Industry Certifications: Career Essentials in Generative AI (Microsoft & LinkedIn Learning, 2024) | Data Analysis Job Simulation (Deloitte) | MVP to AI-Powered Apps for Modern Web Developers

Leadership & Campus Roles: Video Editing Lead, Google Developer Groups (GDG), CIT (2025-2026) | Event Chairperson, Rotaract Club, CIT (2026)